

Tokenization Note

Definition

Tokenization in NLP is the process of transforming raw text into a structured sequence of tokens (words, sub-words, or characters), which are then mapped to numerical identifiers from a fixed vocabulary so that they can be processed by machine learning models.

Types

i) Word-level Tokenization

Each word = one token.

Problem: Out-of-Vocabulary (OOV).

Example: "unbelievable" → ["unbelievable"]

ii) Character-level Tokenization

Each character = one token.

Problem: Very long sequences.

Example: "unbelievable" → ["u", "n", "b", "e", "l", "i", "..."]

iii) Subword-level Tokenization (modern standard)

Breaks words into frequent sub-parts (balance between word & character).

Example: "unbelievable" → ["un", "believe", "able"]

Used in BERT (WordPiece), GPT (BPE), T5 (Unigram).

Key Concepts I Learned

Tokenization vs Encoding vs Embedding

Tokenization = text → tokens.

Encoding = tokens → IDs.

Embedding = IDs → dense vectors (meaningful number representations).

Vocabulary Role

Tokenizer has a fixed dictionary. Unknown words are split into smaller known units.

Example: "acetylsalicylic" (medical word) might break into many sub-words.

Special Tokens

Models add extra tokens like [CLS] (start), [SEP] (separator), [PAD] (padding), [UNK] (unknown). These help the model understand structure, separate inputs, and ignore empty parts.

Attention Mask & Padding

Sentences have different lengths, but models want equal length.

Short sentences are padded with [PAD], and attention masks tell the model to ignore them.

Why Sub-word Tokenization Exists

Word-level → huge vocab, OOV(Out-of-vocabulary) problem.

Character-level → too long, loses meaning.

Subword-level → best balance, handles new words, keeps meaning.